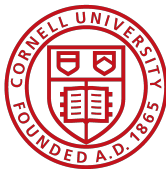


Merton's Credit Default Risk Framework

Application of Black–Scholes–Merton Model to Credit Risk

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Background and Introduction

Credit risk in debt markets

Finance rests on promises to repay. Credit risk permeates large debt markets that shape capital allocation, balance sheets, and macro-financial stability.

- U.S. Treasuries ($\sim \$28.3$ T, $\sim 60\%$)
- U.S. corporate bonds ($\sim \$11.2$ T, $\sim 24\%$)

Challenge: quantifying default risk

Lenders must **judge whether a counterparty can and will repay debt under stress**. However, outcomes are uncertain, path-dependent, and sensitive to macro conditions, collateral, and more. Mis-measured credit risk exposure can turn small shocks into huge losses (e.g. 2008 Financial Crisis).

Motivation: Why stochastic calculus?

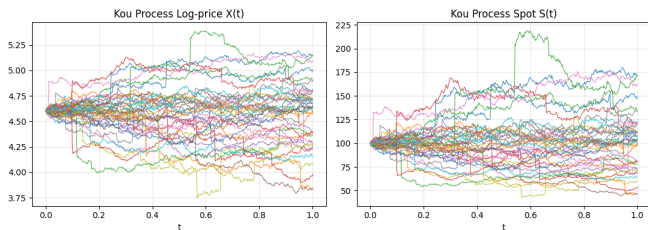


Figure: Kou's jump-diffusion process

- Model uncertain cash flows and default timing (diffusions, jump-diffusions; first passage / hazard jumps).
- Support arbitrage-free pricing, path-dependent hedging, and dynamic risk measures.

Definition of Default: Economic setup (Merton 1974)

Fundamental Accounting Equation

Let A_t = market value of firm assets (unlevered firm value).

Claims split between equity E_t and a **single** zero-coupon debt B_t promising fixed payment D at maturity T .

$$A_t = E_t + B_t$$

where B_t is the *market value of debt* at time t (not face value).

Definition of default at maturity T

$$\text{default at } T \iff E_T \leq 0 \iff A_T \leq D.$$

- Default if promised principal D exceeds available assets A_T .
- Equity wiped out; debt holders absorb shortfall ($\min(A_T, D)$).

Credit Valuation

European Style Option Payoff

(Accounting Identity) $A_t = E_t + B_t$, pay D at maturity T

$$\textbf{(Equity)} \quad E_T = (A_T - D)^+ = \begin{cases} A_T - D, & A_T > D \quad (\text{Healthy}), \\ 0, & A_T \leq D \quad (\text{Default}), \end{cases}$$

$$\textbf{(Debt)} \quad B_T = \min(A_T, D) = \begin{cases} D, & A_T > D \quad (\text{Healthy}), \\ A_T, & A_T \leq D \quad (\text{Default}). \end{cases}$$

- Shareholders keep residual $A_T - D$ if firm is healthy or receive nothing in default (limited liability).
- Debt value is either D if healthy or A_T in default (liquidation).
- Equivalently, debt is a **long** claim to D and a **short** put on firm assets (risky debt = safe debt – put : $D - (D - A_T)^+$).

Merton's Credit Default Risk Framework

BSM Pricing Model: Equity Valuation

Today's values are discounted risk-neutral expectations of maturity payoffs. Equity value with BS call/put prices $c^{\text{BS}}(\cdot)$, $p^{\text{BS}}(\cdot)$ on $S_t = A_t$ following GBM, strike $K = D$:

$$E_t = \mathbb{E}^{\mathbb{Q}}[e^{-r(T-t)}(A_T - D)^+ | \mathcal{F}_t] = c^{\text{BS}}(S_t = A_t, K = D, T, \sigma, r)$$

Closed form (Black–Scholes–Merton)

$$c^{\text{BS}} = A_t \Phi(d_1) - De^{-r(T-t)}\Phi(d_2)$$

Φ stands for the CDF of Gaussian distribution

$$d_1 = \frac{\ln(A_t/D) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t}.$$

Merton's Default Risk Framework

BSM Pricing Model: Debt Valuation

Put-call parity

$$p_t^{\text{BS}} = c_t^{\text{BS}} + D e^{-r(T-t)} - A_t$$

$$\Rightarrow B_t = D e^{-r(T-t)} - p^{\text{BS}}(S_t = A_t, K = D, T, \sigma, r).$$

where

$$p^{\text{BS}} = D e^{-r(T-t)} \Phi(-d_2) - A_t \Phi(-d_1),$$

$$d_1 = \frac{\ln(A_t/D) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t}.$$

Derivation of Black–Scholes–Merton Model: Call Option

1. **GBM under $\mathbb{Q}$:** $dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}}$ ($S_t = A_t$ in Merton).
2. **Itô's Lemma:** for $f(S_t, t)$ be the price of a European claim with maturity payoff $g(S_T)$,

$$df = \left(\partial_t f + rS \partial_S f + \frac{1}{2} \sigma^2 S^2 \partial_{SS} f \right) dt + \sigma S \partial_S f dW_t^{\mathbb{Q}}.$$

$$(\text{Discounted price}) \quad X_t = e^{-rt} f_t, \quad dX_t = e^{-rt} df_t - re^{-rt} f_t dt.$$

3. **Martingale:** X_t is $\mathbb{Q}$ -martingale (Arbitrage-free) $\Rightarrow dt$ -term = 0

$$(\text{Drift-less}) \quad \partial_t f + rS \partial_S f + \frac{1}{2} \sigma^2 S^2 \partial_{SS} f - rf = 0, \quad f(S, T) = g(S).$$

4. **Feynman–Kac:**

$$f(S_t, t) = \mathbb{E}^{\mathbb{Q}}[e^{-r(T-t)} g(S_T) \mid \mathcal{F}_t]; \quad g(S) = (S - K)^+$$

5. **Close-form:** $c^{\text{BS}} = S_t \Phi(d_1) - Ke^{-r(T-t)} \Phi(d_2)$



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Merton, R. C. (1974)

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Lando, D., 2004

Thank you for your time!

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Source: [https://github.com/restinghouse0203/
Merton-Credit-Default-Risk-Model](https://github.com/restinghouse0203/Merton-Credit-Default-Risk-Model)